

# HW 1

## Problem 1

c)

For  $n > 3$

$$D_n = \{e, x, x^2, \dots, x^{n-1}, y, yx, yx^2, \dots, yx^{n-1}\}$$

$$x = \frac{360^\circ}{n}$$

**Claim:** For all  $1 \leq j < n$  we have  $yx^j = x^{n-j}y$ .

*Proof.* Base Case:  $m = 1 \implies yx = x^{n-1}y \checkmark$

Inductive Step: Assume for  $k \geq 1$  the statement is true

Consider:

$$\begin{aligned} yx^{k+1} &= yx^k x && \text{Apply inductive hypothesis on } yx^k \\ &= x^{n-k} yx \\ &= x^{n-k} x^{n-1} y \\ &= x^n x^{n-(k+1)} y \\ &= x^{n-(k+1)} y \end{aligned}$$

□

This  $\implies x^m y = yx^{n-m}$  also

Consider:

$$\begin{aligned} R &= \begin{pmatrix} \cos\left(\frac{360^\circ}{n}\right) & -\sin\left(\frac{360^\circ}{n}\right) \\ \sin\left(\frac{360^\circ}{n}\right) & \cos\left(\frac{360^\circ}{n}\right) \end{pmatrix} \\ S &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ SR^j &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & -\sin\left(\frac{360^\circ j}{n}\right) \\ \sin\left(\frac{360^\circ j}{n}\right) & \cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix} = \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & -\sin\left(\frac{360^\circ j}{n}\right) \\ -\sin\left(\frac{360^\circ j}{n}\right) & -\cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix} \end{aligned}$$

Let  $0 \leq j < n$

$$R^{-j}S = \begin{pmatrix} \cos\left(-\frac{360^\circ j}{n}\right) & -\sin\left(-\frac{360^\circ j}{n}\right) \\ \sin\left(-\frac{360^\circ j}{n}\right) & \cos\left(-\frac{360^\circ j}{n}\right) \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Since  $\cos(-\theta) = \cos(\theta)$  and  $\sin(-\theta) = -\sin(\theta)$ , this becomes:

$$R^{-j}S = \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & \sin\left(\frac{360^\circ j}{n}\right) \\ -\sin\left(\frac{360^\circ j}{n}\right) & -\cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix}$$

The matrix multiplication gives:

$$= \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & -\sin\left(\frac{360^\circ j}{n}\right) \\ -\sin\left(\frac{360^\circ j}{n}\right) & -\cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix}$$

which is the same as the left-hand side.

Thus, we have shown that:

$$SR^j = R^{-j}S$$

To show  $R^{-j} = R^{n-j}$ , consider: Let  $0 \leq j < n$

$$R^{-j} = \begin{pmatrix} \cos\left(-\frac{360^\circ j}{n}\right) & -\sin\left(-\frac{360^\circ j}{n}\right) \\ \sin\left(-\frac{360^\circ j}{n}\right) & \cos\left(-\frac{360^\circ j}{n}\right) \end{pmatrix}$$

$$R^{n-j} = \begin{pmatrix} \cos\left(\frac{360^\circ(n-j)}{n}\right) & -\sin\left(\frac{360^\circ(n-j)}{n}\right) \\ \sin\left(\frac{360^\circ(n-j)}{n}\right) & \cos\left(\frac{360^\circ(n-j)}{n}\right) \end{pmatrix}$$

Noting that:

$$\cos\left(\frac{360^\circ(n-j)}{n}\right) = \cos\left(\frac{360^\circ n}{n} - \frac{360^\circ j}{n}\right) = \cos\left(-\frac{360^\circ j}{n}\right)$$

$$\sin\left(\frac{360^\circ(n-j)}{n}\right) = \sin\left(\frac{360^\circ n}{n} - \frac{360^\circ j}{n}\right) = -\sin\left(\frac{360^\circ j}{n}\right)$$

we have:

$$R^{n-j} = R^{-j}$$

Subgroup of  $GL_2(\mathbb{R})$  :  $\langle R, S \mid R^n = S^2 = I, SR = R^{n-1}S \rangle = A$

- (Associativity) Given by matrix multiplication
- (Identity)

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

this is a zero-degree rotation

- (Inverses) Each rotation matrix  $R^k$  s.t.  $1 \leq k < n$  has an inverse  $R^{n-k}$ . Each reflection matrix  $SR^j$  s.t.  $0 \leq j < n$  has an inverse itself.

$$R^k R^{n-k} = R^n = I = R^n = R^{n-k} R^k$$

$$SR^j SR^j = SR^j R^{n-j} S = SIS = SS = I$$

- (Closure)

1. Let  $R^j$  and  $R^k$  s.t.  $1 \leq j, k < n$

$$R^j R^k = \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & -\sin\left(\frac{360^\circ j}{n}\right) \\ \sin\left(\frac{360^\circ j}{n}\right) & \cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix} \begin{pmatrix} \cos\left(\frac{360^\circ k}{n}\right) & -\sin\left(\frac{360^\circ k}{n}\right) \\ \sin\left(\frac{360^\circ k}{n}\right) & \cos\left(\frac{360^\circ k}{n}\right) \end{pmatrix}$$

Using the angle addition formulas  $\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$  and  $\sin(a+b) = \sin(a)\cos(b) + \cos(a)\sin(b)$

$$= \begin{pmatrix} \cos\left(\frac{360^\circ j+k}{n}\right) & -\sin\left(\frac{360^\circ j+k}{n}\right) \\ \sin\left(\frac{360^\circ j+k}{n}\right) & \cos\left(\frac{360^\circ j+k}{n}\right) \end{pmatrix}$$

This is another rotation matrix specifically  $R^{j+k \bmod n}$

2. Let  $SR^j$  and  $R^k$  be a reflection and a rotation s.t.  $0 \leq j < n$  and  $1 \leq k < n$ . Then

$$(SR^j)R^k = SR^{j+k \bmod n}$$

This is another reflection in the group.

$$R^k(SR^j) = (R^kS)R^j = S(R^{n-k}R^j) = SR^{n+j-k \bmod n}$$

This is another reflection in the group.

3. Let  $SR^j$  and  $SR^k$  be reflections s.t.  $0 \leq j, k < n$

$$SR^jSR^k = R^{n-j}SSR^k = R^{n+k-j \bmod n}$$

This is another rotation in the group.

Let  $\varphi : D_n \mapsto A$  s.t.  $\varphi(x^j) = R^j$ , and  $\varphi(yx^j) = SR^j$  s.t.  $0 \leq j < n$

Well Defined: Le

Homomorphism: Let  $a, b \in D_n$

1. If  $a = x^j$  and  $b = x^k$  s.t.  $1 \leq j, k \leq n$

$$\varphi(x^jx^k) = \varphi(x^{j+k}) = R^{j+k} = R^jR^k = \varphi(x^j)\varphi(x^k)$$

2. If  $a = x^j$  and  $b = yx^k$  s.t.  $0 \leq j, k < n$

$$\varphi(x^jyx^k) = \varphi(yx^{n-j}x^k) = SR^{n+k-j} = R^jSR^k = \varphi(x^j)\varphi(yx^k)$$

$$\varphi(yx^kx^j) = \varphi(yx^{k+j}) = SR^{k+j} = SR^kR^j = \varphi(yx^k)\varphi(x^j)$$

3. If  $a = yx^j$  and  $b = yx^k$  s.t.  $0 \leq j, k < n$

$$\varphi(yx^jyx^k) = \varphi(x^{n-j}yyx^k) = \varphi(x^{n+k-j}) = R^{n+k-j} = R^{n-j}SSR^k = SR^jSR^k = \varphi(yx^j)\varphi(yx^k)$$

Injective: Let  $a, b \in D_n$  s.t.  $\varphi(a) = \varphi(b)$

Small Proof Before: Let  $G, H$  be groups and  $\varphi$  be a homomorphism from  $G$  to  $H$  then

$$\varphi(g)^{-1} = \varphi(g^{-1}) \forall g \in G$$

*Proof.* Let  $g \in G$

$$e_H \varphi(e_G) = e_H \varphi(e_G e_G) = e_H \varphi(e_G) \varphi(e_G) \implies e_H = \varphi(e_G)$$

$$e_H = \varphi(e_G) = \varphi(gg^{-1}) = \varphi(g) \varphi(g^{-1}) \implies \varphi(g)^{-1} = \varphi(g^{-1})$$

□

$$\begin{aligned}\varphi(a) &= \varphi(b) \\ \varphi(a) \varphi(b)^{-1} &= I \\ \varphi(ab^{-1}) &= I\end{aligned}$$

Cases:

1. If  $a = x^j$  and  $b = x^k$  for  $0 \leq j, k < n$

$$I = \varphi(x^j x^{-k}) = R^j R^{-k} \implies j = k$$

Therefore  $a = b$

2. If  $a = yx^j$  and  $b = yx^k$  for  $0 \leq j, k < n$

$$I = \varphi(yx^j yx^k) = SR^j SR^k = R^{n-j} SSR^k = R^{n+k-j} \implies k = j$$

Therefore  $a = b$

3. If  $a = x^j$  and  $b = yx^k$  for  $0 \leq j, k < n$

$$I = \varphi(x^j yx^k) = R^j SR^k = SR^{n+k-j} \implies R^{j-n-k} = S \Rightarrow \Leftarrow$$

This is a contradiction since this would turn  $A$  into only having  $R$

Therefore  $a = b$  thus  $\varphi$  is Injective

Surjective:

Let  $a \in A$

Cases:

1. If  $a = R^k$  s.t.  $0 \leq k < n$  Consider  $\varphi(x^k) = R^k = a$
2. IF  $a = SR^k$  s.t.  $0 \leq k < n$  Consider  $\varphi(yx^k) = SR^k = a$

Therefore  $\varphi$  is surjective

Therefore  $\varphi$  is bijective

Therefore  $\varphi$  is an isomorphism

d)

Let  $G$  be a group Consider  $Z(G)$

1. (Identity) Consider  $e, g \in G$ .  
 $eg = ge \implies e \in Z(G)$
2. (Closure) Let  $z, h \in Z(G)$  and  $g \in G$   
 $zhg = zgh = gzh \implies zh \in Z(G)$
3. (Inverses) Let  $z \in Z(G)$  and  $g \in G$   
 $zg = gz \implies g^{-1}z^{-1} = z^{-1}g^{-1} \implies z^{-1} \in Z(G)$

Therefore  $Z(G)$  is a subset of  $G$

- If  $n$  is odd then  $Z(G) = \{e\}$
- If  $n$  is even then  $Z(G) = \{e, x^{n/2}\}$

Proven in c)  $x^j y = y x^{n-j}$  for  $1 \leq j < n$

Thus only  $x^j$  commutes with  $y$  if  $j = n - j \implies j = n/2$

and  $x^j$  commutes with all  $x^k$  since associativity and that  $x^k x^j = x^{k+j}$

and since  $x^j$  commutes with  $x^k$  and  $y$  it will commute with all reflections

Therefore this is only possible if  $n$  is even.

Geometrically:

When  $n$  is odd this is just the do nothing to the  $n$ -gon.

When  $n$  is even this is just a  $180^\circ$  rotation.

e)

Let  $G, H$  be a group and  $\varphi$  be a homomorphism between them and let  $g \in G \forall n \in \mathbb{Z}, \varphi(g^n) = [\varphi(g)]^n$

*Proof.* Base Case:  $n = 0 \quad \varphi(g^0) = \varphi(e_1) = e_2 = [\varphi(g)]^0$

Assume statement is true for  $k \geq 0$  for  $k \in \mathbb{N}$

Consider:  $k + 1$

$$\varphi(g^{k+1}) = \varphi(g)\varphi(g^k) = \varphi(g)[\varphi(g)]^k = [\varphi(g)]^{k+1}$$

$$\text{Note: } e_2 = \varphi(e_1) = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1}) \implies [\varphi(g)]^{-1} = \varphi(g^{-1})$$

This proves the statement for all negative integers since we just apply the induction on  $g$  then find its inverse □

If  $\varphi$  is an isomorphism then  $\text{ord}(g) = \text{ord}(\varphi(g))$

*Proof.* Assume  $g \in G$  s.t.  $\text{ord}(g) = n \in \mathbb{N}$

$$\implies \varphi(g^n) = \varphi(e_G) = e_H \iff \text{a proof in c)}$$

$$\implies \varphi(g)^n = e_H \implies \text{ord}(\varphi(g)) \text{ divides } n$$

Since  $\varphi$  is injective, if  $\varphi(g)^m = e_H$  for some  $m < n$  then  $g^m = e_G$  but this contradicts that  $g$  has order  $n$

$$\text{Thus } \text{ord}(\varphi(g)) = \text{ord}(g) \quad \square$$

Thus, by the proof above an isomorphism has 2 choices for where  $x$  goes either  $\tau$  or  $\tau^2$  and once you pick  $x$  there is only one place for  $x^2$

and there is 1 choice for  $e$

and  $y$  has 3 choices  $\sigma, \sigma\tau, \sigma\tau^2$ , and since the isomorphism has to preserve group structure there is only one choice for  $yx$  and  $yx^2$  after you pick  $y$

Thus, there are in total 6 isomorphisms.

They are:

- |                                                                                                                                           |                                                                                                                                           |                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    • $e$ to $e$<br>• $x$ to $\tau$<br>• $x^2$ to $\tau^2$<br>• $y$ to $\sigma$<br>• $yx$ to $\sigma\tau$<br>• $yx^2$ to $\sigma\tau^2$ | 3.    • $e$ to $e$<br>• $x$ to $\tau^2$<br>• $x^2$ to $\tau$<br>• $y$ to $\sigma\tau^2$<br>• $yx$ to $\sigma\tau$<br>• $yx^2$ to $\sigma$ | 5.    • $e$ to $e$<br>• $x$ to $\tau$<br>• $x^2$ to $\tau^2$<br>• $y$ to $\sigma\tau^2$<br>• $yx$ to $\sigma$<br>• $yx^2$ to $\sigma\tau$ |
| 2.    • $e$ to $e$<br>• $x$ to $\tau^2$<br>• $x^2$ to $\tau$<br>• $y$ to $\sigma\tau$<br>• $yx$ to $\sigma$<br>• $yx^2$ to $\sigma\tau^2$ | 4.    • $e$ to $e$<br>• $x$ to $\tau^2$<br>• $x^2$ to $\tau$<br>• $y$ to $\sigma$<br>• $yx$ to $\sigma\tau^2$<br>• $yx^2$ to $\sigma\tau$ | 6.    • $e$ to $e$<br>• $x$ to $\tau$<br>• $x^2$ to $\tau^2$<br>• $y$ to $\sigma\tau$<br>• $yx$ to $\sigma\tau^2$<br>• $yx^2$ to $\sigma$ |

Since all of these mappings are injective they are bijective since  $|D_3| = |S_3|$ .  
 homomorphisms

1.    •



## **Problem 2**

**a)**

### Problem 3

Separate Proof:

*Proof.* Let  $G$  be a finite group and  $g \in G$  and let  $\text{ord}(g) = n \in \mathbb{N}$

Consider  $\langle g \rangle$ :

1.  $g^n = e \in \langle g \rangle$
2. Let  $g_1, g_2 \in \langle g \rangle \implies g_1 = g^j$  and  $g_2 = g^k$  for  $j, k \in \mathbb{N}_{\leq n}$  thus  $g_1 g_2 = g^j g^k = g^{j+k \bmod n} \in \langle g \rangle$
3. Let  $g_1 \in \langle g \rangle \implies g_1 = g^j$  for  $j \in \mathbb{N}_{\leq n}$

Consider:  $g^{n-j} \in \langle g \rangle$

$$g^{n-j} g^j = e = g^j g^{n-j} \implies g_1^{-1} = g^{n-j} \in \langle g \rangle$$

Thus for all finite groups  $G$  and  $g \in G$  with finite order,  $\langle g \rangle$  is a subgroup of  $G$

$\implies$  by Lagrange's Theorem  $\text{ord}(g)$  divides  $\text{ord}(G)$  □

Another Separate proof:

*Proof.* Set  $m = [G : H]$  and take  $g_1, g_2, \dots, g_m \in G$  so that  $g_1 H, g_2 H, \dots, g_m H$  are all distinct.

Also,  $|g_i H| = |H| \ \forall 1 \leq i \leq m$ . This is because you take all elements of  $H$  and add  $g_i$  to them.

Since  $G = g_1 H \cup g_2 H \cup \dots \cup g_m H$  and these are all disjoint by Lagrange's Theorem

By the multiplication principle,  $|G| = m|H| = [G : H]|H|$  □

*Proof.* Let  $p$  be prime and  $G$  be a group of order  $2p$

1. Assume  $p = 2$

Thus  $G$  has 4 elements

- (a) Assume  $G$  has an element with order 4

$\implies \langle g \rangle$  has 4 elements since  $G$  also has 4 elements and  $\langle g \rangle \subseteq G$  thus there is no element in  $G$  that is not in  $\langle g \rangle$  thus  $G = \langle g \rangle$  thus  $G \cong C_4$

- (b) Assume  $G$  has no elements with order 4

Thus by my proof above all elements have order 2 for all non-identity elements

Thus  $G = e, a, b, c$  then  $a^2 = b^2 = c^2 = e$ . WLOG let  $c = ab$

Consider  $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$

Let  $\varphi : G \mapsto D_2$  s.t.  $\varphi(e) = e, \varphi(a) = x, \varphi(b) = y, \varphi(ab) = yx = xy = \varphi(ba)$

This is a well-defined function because every input has one output

Injective:

This is injective by definition.

Since this is injective and  $G$  and  $D_2$  have 4 elements this makes  $\varphi$  surjective. Thus it is bijective.

Homomorphism:

$$\varphi(ab) = xy = \varphi(a)\varphi(b)$$

$$\varphi(ba) = yx = \varphi(b)\varphi(a)$$

Let  $g \in G$

$$\varphi(eg) = \varphi(g) = \varphi(e)\varphi(g)$$

$$\varphi(ge) = \varphi(g) = \varphi(g)\varphi(e)$$

Thus  $\varphi$  is an isomorphism thus  $G \cong D_2$

2. Assume  $G$  has an element  $g$  s.t.  $\text{ord}(g) = 2p$  for odd primes

$\implies \langle g \rangle$  has  $2p$  elements since  $G$  also has  $2p$  elements and  $\langle g \rangle \subseteq G$  thus there is no element in  $G$  that is not in  $\langle g \rangle$  thus  $G = \langle g \rangle$  thus  $G \cong C_{2p}$

3. Assume  $G$  has no elements of order  $2p$  and  $p$  is odd

For contradiction assume  $G$  has no elements of order  $p \implies$  by my proof above that all non-identity elements have order of 2

Take  $e \neq a \neq b \neq e \in G$

Consider  $\{e, a, b, ab\} \leq G \implies$  by Lagrange's Theorem that 4 divides  $2p \implies 2$  divides  $p \implies \Leftarrow$

Thus  $G$  has an element  $g \in G$  with order  $p$ . Take  $\langle h \rangle = H$  and proven above  $H$  is a subgroup of  $G$ . By proof above  $[G : H] = \frac{|G|}{|H|} = \frac{2p}{p} = 2$ . Take  $g \notin H \implies G = H \sqcup gH$

Consider:  $g^2H$

- (a) Assume  $g^2H = H \implies g^2 \in H$

$\implies \text{ord}(g^2)$  divides  $p$  by Lagrange's Theorem  $\implies \text{ord}(g^2) = 1$  or  $\text{ord}(g^2) = p$

- i. If  $\text{ord}(g^2) = 1 \implies g^2 = e \implies \text{ord}(g)$  divides 2  $\implies \text{ord}(g) = 2$  or 1

A. If  $\text{ord}(g) = 1$  then  $\langle g \rangle = \langle g^2 \rangle = \{e\}$  which is a subset of  $H$  which  $\implies g \in H$  this is a contradiction

- ii. If  $\text{ord}(g^2) = p \implies g^{2p} = e \implies \text{ord}(g)$  divides  $2p$  and since there is no elements of order  $2p$

$\implies \text{ord}(g) = 1$  or  $p$

A. If  $\text{ord}(g) = p$  then  $\langle g \rangle = \langle g^2 \rangle$  which is a subset of  $H$  which  $\implies g \in H$  this is a contradiction

B. If  $\text{ord}(g) = 1$  then  $g = e$  but  $g^2 \neq e$  since  $\text{ord}(g^2) = p$  this is a contradiction

Therefore  $\text{ord}(g) = 2$  by 3.(a).i

$$(b) \ g^2H = gH$$

Take  $h, j \in H$  s.t.  $g^2h = gj \implies g = jh^{-1} \in H$  this is a contradiction

Therefore anything not in  $H$  has order 2.

$$gh \in gH \implies gh \notin H \implies \text{ord}(gh) = 2$$

$e = (gh)^2 = ghgh \implies h^{-1}g^{-1} = gh \implies gh = h^{p-1}g \implies gh^k = h^{p-k}g$  Since Cosets are a partition

$$G = H \sqcup gH = \{e, h, h^2, \dots, h^{p-1}, g, gh, \dots, gh^{p-1}\} = \langle h, g \mid e = h^p = g^2, gh = h^{p-1}g \rangle = D_p$$

□